COMPREHENSIVE RESEARCH REPORT

AI, Mobility and Smart Cities
Optimizing Traffic Signals with Mobility Data and AI

Team B — 2026 Crimson Silicon Valley SW & AI Leaders Program
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Research workstreams
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Document purpose: Consolidated research findings, verified/attributed case studies, algorithmic and engineering theory, datasets, implementation references, methodological limitations, and division of research responsibilities. This is a research report, not a funding proposal, and does not claim completed experiments.

Evidence legend: [F] reported field deployment; [R] academic research or theory; [V] vendor/organization claim; [D] proposed Team B design; [U] requires independent source/method verification.

Executive research synthesis

The original Team B activity report (13 October 2026) framed three core questions: bias when navigation-app users are not representative of all road users; congestion spillover when a single intersection is optimized without considering neighboring links; and credible effectiveness evaluation after accounting for weather, season, weekday, roadworks, and traffic-demand changes [1]. This report preserves those questions and expands the investigation into traffic engineering, data science, and AI control.

The evidence indicates that signal optimization is not synonymous with deep learning. Engineer-reviewed data recommendations (Google Project Green Light), decentralized schedule-driven adaptation (SURTRAC), mature adaptive systems (SCOOT/SCATS), queue-pressure control, MPC, and multi-agent RL all address related but distinct operational problems. Cross-study percentages are not directly comparable because networks, demand, baselines, metrics and evaluation periods differ [2–11].

Research conclusion: A credible AI evaluation must include strong conventional baselines, physically valid downstream receiving capacity, a realistic observation model, enforceable phase safety, and matched, repeated demand scenarios. The proposed four-junction SUMO corridor is an engineering testbed for these questions, not an empirical representation of any city [1,12–15].

1. Research scope, questions and evidence protocol

1.1 Questions inherited from the original activity report

- Representativeness: What is missed when signal optimization relies on vehicles using navigation or connected-vehicle services, rather than all traffic, buses and pedestrians? [1]
- Spillover: How should adjacent intersections exchange information to avoid transferring congestion downstream? [1]
- Causal evaluation: How can before/after improvements be separated from weather, seasonal demand, weekdays, incidents, and construction? [1]
- Safety and governance: Why use engineer-reviewed recommendations rather than fully automated signal changes; what fallback is appropriate when sensors or AI fail? [1]

1.2 Evidence quality rules

Evidence class	Interpretation	Reporting requirement
Field deployment / independent evaluation	Potentially relevant operational evidence	Name evaluator, baseline, study period, comparison groups and confounders.
Researcher-led field study	Real-world observation, potential developer involvement	Attribute results and distinguish independent replication.
Simulation experiment	Evidence conditional on simulated assumptions	Disclose network, demand, seeds, calibration and baselines.
Vendor or municipal announcement	Useful deployment claim, not automatically causal	Identify reporting organization and missing raw data.
Proposed Team B parameters	Engineering hypotheses/design decisions	Never present as measured field conditions.

2. Real-world systems and case studies

2.1 Google Project Green Light: engineer-reviewed recommendations

Project Green Light analyzes aggregated driving patterns to recommend signal timing changes; city traffic engineers review and implement changes using existing infrastructure. The original Team B report cites Google's May 2025 description of operation across 18 cities on four continents and Google-estimated potential reductions of up to 30% in stops and up to 10% in intersection emissions. Those potential percentages are organizational estimates, not independently transferable outcomes [1,2].

Boston reported recommendations applied at 114 intersections across 20 neighborhoods. Its announcement reports average reductions of 20% in unnecessary stops and 13.5% in delay in the evaluated phase, with INRIX involved in analysis. The public statement does not provide a fully reconstructable causal research dataset, so these are city-reported field results, not established causal effects for other networks [1,3].

Engineering relevance: Offline recommendations with human review reduce the control-system integration burden and create a plausible first commercialization path. Research limitation: sampling bias and selection of retimed intersections need independent examination.

2.2 SURTRAC: decentralized predictive coordination

SURTRAC uses local traffic observations to construct schedules of approaching vehicle clusters and communicates predicted outflows to downstream intersections. The Pittsburgh nine-intersection pilot reported substantial improvements in travel time and waiting time relative to the preceding control scheme. The published pilot is associated with the developing research group; its findings should not be characterized as fully independent validation [4].

Published descriptions of the pilot report greater than 31% reduction in travel time and greater than 40% reduction in waiting time under the studied evaluation. These figures should not be blended with different summaries or network phases. Research relevance: SURTRAC demonstrates that decentralized prediction and communication can provide coordination without a monolithic centralized controller [4].

2.3 Jeju Aejo-ro: digital-twin-assisted signal improvement

The original Team B report describes a 13.4 km Aejo-ro section between Hagwi 1-ri and Dongsaem intersections. It attributes signal timing adjustments to smart-intersection observations and digital-twin simulations, with reported travel time decreasing from 23 min 21 s to 19 min 31 s, average speed increasing from 37.6 to 42.9 km/h, and delay decreasing by 31.3% [1,5].

These numbers are preserved as source-attributed local reporting [U]. A primary technical report detailing sample size, measurement windows, weather controls, detector methodology and uncertainty was not available in the collected materials. No causal inference is made here.

2.4 Busan: video sensing and adaptive signal allocation

The original report describes AI analysis of smart-intersection video and real-time directional signal allocation. It reports a mean speed increase of approximately 2.25 km/h and intersection delay decreasing from 13.9 to 12.5 seconds (approximately 10.1%). It also notes the sharing of real-time signal information with navigation services [1,6].

These figures remain municipal/media-reported [U] pending retrieval of the underlying evaluation. The case is useful for studying sensor recognition errors, autonomous versus engineer-reviewed operation, and fallback logic.

2.5 SCOOT and SCATS: mature adaptive comparators

SCOOT and SCATS are established adaptive signal control families that adjust traffic timing using detector information and operational rules. Their existence is important: an RL model should not be described as introducing adaptation itself. Supplier-reported typical performance gains vary by deployment and should not be interpreted as a universal benchmark [7,8].

2.6 Comparative synthesis

System	Decision mode	Evidence caveat	Question for Team B
Green Light	Data-informed recommendations; engineer review	City/vendor reported; methods partly opaque	Does biased navigation data still yield useful timing recommendations?
SURTRAC	Local predictive scheduling + neighbor messages	Developer-led field study	Can distributed coordination reduce spillback?
Jeju	Sensor + digital twin + timing adjustment	Technical evaluation not retrieved	How to validate simulation-to-field transfer?
Busan	Video-driven adaptive operation	Primary evaluation not retrieved	How to fail safely on perception error?
SCOOT / SCATS	Mature detector-driven adaptation	Vendor outcomes are context dependent	Does AI beat a strong conventional controller?

3. Traffic engineering theory

3.1 Flow, density and speed

$$q = k \times v$$

Flow q (vehicles/hour) equals density k (vehicles/km) multiplied by space-mean speed v (km/hour) under consistent aggregation. High speed does not necessarily imply high throughput; congested links may exhibit both low speed and low discharge [9,10].

3.2 Saturation flow, capacity and degree of saturation

$$s = 3600 / h_s \; ; \; c_i = s_i \times g_i / C \; ; \; X_i = v_i / c_i$$

Here h_s is saturation discharge headway (seconds/vehicle), s_i saturation flow (veh/h of effective green), g_i effective green, C cycle length, and X_i volume-to-capacity ratio. Example [D]: h_s = 2 s gives s = 1800 veh/h; g = 40 s and C = 90 s give nominal capacity 800 veh/h. At demand 700 veh/h, X = 0.875. These are illustrative calculations, not calibrated observations [9,10].

3.3 Queue conservation and downstream receiving capacity

$$Q_i(t+\Delta t) = Q_i(t) + A_i(t) - D_i(t)$$

$$D_i(t) \le \min[Q_i(t)+A_i(t),\, S_i(t),\, R_j(t)]$$

Arrivals and departures must be counted over the same interval; signal service S_i and downstream receiving allowance R_j are also expressed in compatible vehicle counts. Movement-specific turning flows and shared receiving capacity must be enforced to avoid double-counting. A local queue reduction can simply relocate congestion into the next link [9–11].

3.4 Finite link storage and spillback

$$N_{storage} \approx \text{floor}(L_{usable} / l_{effective})$$

For a 300 m one-lane link and 7.5 m effective stopped spacing, nominal storage is approximately 40 passenger vehicles [D]. This ignores intersection setbacks and heterogeneous vehicle lengths. Distinguish a 90%-storage warning from actual downstream blockage preventing upstream discharge; record both near-capacity duration and blocking duration.

3.5 Signal phases, lost time and coordination

Effective green and displayed green are not interchangeable. Phase transitions must include clearance intervals and any pedestrian service. A preliminary 90 s cycle [D] can use 48 s east–west green, 3 s yellow, 1 s all-red, 34 s north–south green, 3 s yellow and 1 s all-red. The clearance values are placeholders; no field safety compliance is implied [9,10,12].

$$t_{travel} = 300 / (50/3.6) = 21.6 \text{ seconds}$$

A one-way progression starting point [D] is offsets [0, 22, 44, 66] seconds for four junctions at 300 m spacing and 50 km/h. Bidirectional progression, platoon dispersion, queues and turn movements can invalidate this simple offset vector. Offset reference conventions must be documented.

4. Proposed four-intersection SUMO research testbed

4.1 Geometry and experimental assumptions

Item	Reference specification [D]
Intersections	J1–J4 at coordinates (0,0), (300,0), (600,0), (900,0) m
Arterial	Two-way east–west; one through lane per direction initially
Cross streets	North–south approaches at each junction; explicit turns in extended model
Speed / internal length	50 km/h; 900 m between first and fourth junction
Simulation	SUMO, Python/TraCI, 1 s step; 900 s warm-up; 3600 s evaluation
Base control	90 s fixed-time reference; actuated and coordinated variants
Progressive fidelity	Through-only proof of correctness → turning traffic → pedestrian phases

4.2 Network generation and signal mapping

Generate nodes, edges, lane connections and signal logic with SUMO tools. Verify each junction’s actual controlled-link indices from the generated network: signal state strings are indexed by controlled connections, not arbitrary compass-direction labels. Use the official traffic-light API and validated programs rather than arbitrary state overrides [12,13].

4.3 Detector layout

Recommended research sensors [D]: upstream induction loops for passage counts; 100–150 m lane-area detectors for occupancy/queue proxies; stop-line observations for discharge headways; and downstream observations for receiving-link congestion. Every sensor must fit within the physical lane length. Distinguish simulated ground truth from the controller’s available observations [12,14].

4.4 Safety supervisor

AI and optimization modules submit proposed actions to an independent supervisor. The supervisor checks compatible greens, phase order, minimum green, required yellow and all-red, applicable maximum green, pedestrian service, and fallback behavior. Invalid actions are rejected and logged. This is simulation-level validation, not traffic hardware certification [9,12,13].

5. Big Data research: sampling, state estimation and evaluation

5.1 Observation models and bias

$$y_t = H_t x_t + \varepsilon_t \; ; \; I_v \sim \text{Bernoulli}(p_v)$$

x_t is the true simulated state; H_t represents sensing and aggregation; ε_t measurement noise; I_v is vehicle-level connected-data participation. A vehicle should retain its participation status across its trip for a penetration-rate experiment. Repeated per-second sampling instead models intermittent reporting, not fleet penetration.

Test uniform coverage at 10%, 25%, 50%, 75% and 100%, then compare approach-biased and speed/state-dependent sampling at matched coverage. The controller must never access hidden full-simulator state except in explicitly labeled oracle experiments.

5.2 Data contracts

Dataset	Essential fields
Vehicle observations	run_id, time_s, pseudonymous_vehicle_id, lane, position_m, speed_mps, observed_flag
Detector observations	run_id, detector_id, interval_start_s, interval_end_s, count, occupancy, missing_flag
Signal log	junction_id, phase_id, requested_action, executed_action, rejection_reason, timestamps
Scenario manifest	network_hash, demand_hash, scenario_id, seeds, turn ratios, sampling mechanism
Outcome table	delay, travel_time, p95, stops, queue, spillback, throughput, unfinished_trips, teleports

5.3 Metrics and interpretation

Primary endpoint: network-wide mean vehicle delay with an explicit population definition. Secondary: mean/p95 travel time, stops/vehicle, maximum and mean queue, actual spillback/blocking duration, throughput, cross-street service, arrivals-on-green, safety violations, runtime and incomplete trips. The original report also included estimated emissions as a secondary metric; emissions estimates require model assumptions and must not be described as measured air quality [1].

$$\text{Improvement (\%)} = 100 \times (\text{baseline metric} - \text{candidate metric}) / \text{baseline metric}$$

The expression applies to lower-is-better outcomes and only when the baseline denominator is positive. Report absolute changes and uncertainty intervals alongside percentages.

5.4 Causal and statistical research

Matched before/after comparisons should align time of day and weekday, and record weather, incidents, season, construction, demand changes and traffic composition. For a simulation experiment, use paired demand seeds, separate calibration/tuning/holdout scenarios, and independent RL training seeds. Estimate confidence intervals for paired differences, report variance and address multiple comparisons. The earlier 20-seed suggestion is a planning target, not a validated power calculation.

5.5 Databricks / scalable geospatial analytics

The activity report identifies a Databricks geospatial workflow using vehicle telematics, GPS, Spark, OpenStreetMap, OSMnx and NetworkX. The industry research questions are ingestion and serving cost at high event rates, spatial joins, and whether analysts can safely deploy model changes through MLOps. This is a candidate architecture and interview agenda, not evidence of measured savings [1,15].

6. AI algorithm research

6.1 Common decision model

$$x_{t+1} = f(x_t, u_t, d_t, w_t)$$

Traffic state x_t evolves under signal decisions u_t, demand d_t and disturbances w_t. The shared objective can penalize normalized delay, queue, stops and spillback; legal signal phases and clearance requirements remain hard constraints rather than mere reward penalties.

6.2 Max-pressure control

$$P_{ij} \approx q_i - q_j \; ; \; P_{phase} = \Sigma \; service_{ij} \times P_{ij}$$

Pressure reflects imbalance between upstream and downstream queues. Practical implementations must use movement-specific queues, turning ratios and service rates, and must respect downstream storage. Varaiya provides theoretical foundations; PressLight uses pressure-informed reinforcement learning. Max-pressure is a strong non-neural benchmark [10,16].

6.3 Deep Q-learning

$$y_t = r_t + \gamma \max_{a'} Q_{target}(s_{t+1}, a')$$

$$L(\theta) = E[(y_t - Q_\theta(s_t, a_t))^2]$$

DQN learns values for discrete legal actions such as phase extension or validated next-phase selection. A centralized joint action space can grow exponentially with junction count; for four agents with four actions each, there are 256 joint combinations. Training stability and action constraints remain significant issues.

6.4 Multi-agent RL and graph attention

Independent RL agents can experience nonstationarity as neighbors learn. Centralized training with decentralized execution (CTDE) can use broader training information while restricting deployed observations. CoLight applies graph attention to learn which neighboring intersections matter in each state. Attention is a learned weighting mechanism, not proof of causality [17].

$$\alpha_{ij} = \exp(e_{ij}) \; / \; \Sigma_{k \in N(i)} \exp(e_{ik})$$

Test independent agents against neighbor-aware agents under balanced demand, directional peaks, turning-ratio shifts, partial observation and downstream blockages.

6.5 Model predictive control

$$\min \Sigma_{k=0 \ldots H-1} \left[q_{t+k}^{T} Q q_{t+k} + u_{t+k}^{T} R u_{t+k} \right]$$

MPC predicts future queue evolution over a finite horizon, optimizes feasible actions, applies the first decision, and repeats after new observations. It offers explicit operational constraints but depends on model quality, forecast reliability and solver latency. A practical first version may select among approved timing plans every 60–120 seconds instead of switching arbitrary phase states.

6.6 Forecasting models

Use historical averages and exponential smoothing as baselines; compare Kalman filtering, GRU/LSTM and graph models only when data volume warrants them. Evaluate both prediction errors (MAE/RMSE) and resulting traffic-control performance. Underestimating a near-spillback queue may be more harmful than a larger error on a low-risk approach.

6.7 Security and privacy

Relevant threat scenarios include forged detector readings, stale/replayed telemetry, model-artifact tampering and unauthorized control commands. Controls include input validation, signed/versioned artifacts, role-based access, audit logs, pseudonymous identifiers, limited retention and safe fallback. The original Tesla visit questions concern vehicle-side traffic-light perception and V2X, which are adjacent to but not evidence of safe roadside signal control [1].

7. Algorithms, software libraries and benchmark datasets

Resource	Role	Limitations / checks
SUMO + TraCI [12,13]	Microscopic network, detectors, signal control	Validate signal indices, routes and vehicle behavior.
SUMO-RL [18]	Gymnasium/PettingZoo RL integration	Check action semantics and safety assumptions.
RESCO [19]	Comparable scenarios and baseline algorithms	Record benchmark versions and network provenance.
LibSignal [20]	Cross-simulator controller framework	Confirm algorithm support for chosen simulator/version.
CityFlow [21]	High-throughput RL traffic simulation	Not directly interchangeable with SUMO.
CoLight repository [17]	Graph-based MARL research implementation	Reproduce settings before transferring claims.
OSM-derived networks [15]	Realistic road topology	Realistic geometry does not imply calibrated demand.

Recommended scenario progression: synthetic corridor first; then independent RESCO-style networks such as Cologne and Ingolstadt for generalization if resources allow. Keep training, validation and holdout demand distinct.

8. Experimental methodology and engineering test specifications

8.1 Controller comparison ladder

Tier	Controller	Research purpose
0	Fixed-time	Sanity check and simple reference.
1	Actuated and optimized coordination	Strong conventional baselines.
2	Max-pressure	Queue-aware non-learning adaptation.
3	MPC	Explicit predictive coordination and constraints.
4	Independent RL and MARL	Learned policies and neighbor coordination.

8.2 Factorial scenario design

Demand: light, medium, near-capacity and oversaturated; directionality: balanced, eastbound peak, westbound peak; observations: complete, detector-only and sampled vehicle data; incidents: blocked downstream link, demand surge, sensor outage; generalization: unseen turn proportions and held-out seeds.

8.3 Acceptance tests

Test family	Examples of required assertions
Network	Exactly four junctions; valid routes; correct lane connections; no unexpected teleports.
Signals	No conflicting protected greens; valid minimum green and clearance; pedestrian timing if present.
Flow	Vehicle conservation; plausible saturation discharge; blocked receiving link limits departures.
Observation	Detector units and timestamps valid; missingness explicit; no oracle leakage.
Safety	Invalid AI proposal rejected; inference timeout triggers safe fallback.
Reproducibility	Same software, inputs and seeds yield matching outputs within documented tolerance.
Evaluation	Include incomplete trips, throughput, queue and spillback, not just completed-trip travel time.

8.4 Experimental integrity

Freeze scenario and metric definitions before final comparisons. Report paired differences and 95% confidence intervals; include RL training-seed variation, computation cost and supervisor interventions. A policy that lowers completed-trip mean delay by stranding vehicles must not be called superior.

9. Calibration, validation and generalization

For synthetic networks, conduct internal plausibility validation against queue conservation, discharge headways, lane-group capacity and finite storage. This is not field calibration. When empirical observations are available, calibrate car-following behavior, turn proportions, volumes, speeds, travel times and queues on one dataset and validate on another [9,12].

$$GEH = \mathrm{sqrt}[2(M - O)^2 / (M + O)]$$

GEH is a diagnostic for comparing modeled and observed traffic volumes when measurement periods are comparable; MAE and RMSE are suitable for speed, queue and travel-time residuals. A single threshold cannot establish comprehensive validity. Calibration assumptions, data quality and uncertainty should be reported.

10. Research limitations and interpretation

- The four-intersection corridor is synthetic until empirical geometry, counts and timing plans are supplied.
- Reported Boston, Jeju and Busan percentage improvements use different baselines and metric definitions and should not be pooled into a meta-analysis.
- Published academic RL improvements may depend on simulator, observation privileges, reward shaping and training seeds.
- Safety constraints in SUMO do not certify a controller for a live traffic cabinet.
- Mobility sampling assumptions may not represent commercial navigation datasets, which have unknown and potentially nonrandom selection processes.
- Travel-time and delay metrics can be biased by incomplete trips; report throughput and unserved demand.

- The Tesla case discussed in the activity report is about vehicle-side perception and safety investigation, not a measured benefit of traffic-signal optimization.

11. Team research division and integration

Member / research lead	Primary responsibility	Research output
Kim Seong Hyeon — Software	SUMO network, TraCI, baseline controllers, safety supervisor, integration and test automation; overseas cases	Executable corridor, APIs, test suite, reproducible runs and demo.
Lee Do Yeong — Big Data	Mobility sampling, detector schema, data processing, causal evaluation and statistical metrics; Databricks questions	Dataset dictionary, bias experiments, KPI pipeline, evidence-quality register.
Lee Yun Seong — AI	Max-pressure/MPC/MARL, rewards, robustness, security/privacy; Jeju/Busan research and Tesla questions	Algorithm review and adapters, training logs, threat model, comparative results.

11.1 Integration contracts

Software defines valid action/phase interfaces and stable experiment orchestration. Big Data defines detector event schemas, scenario manifests, observation masks and outcome metrics. AI uses only permitted observations, proposes feasible actions and records model versions. All three jointly sign off on safety, benchmark fairness, result interpretation and presentation claims.

12. Research questions for external visits and industry interviews

Google

- How is navigation-user sampling bias measured or corrected, especially for buses, pedestrians and underrepresented approaches?
- How are downstream queues, network spillback and adjacent intersections included in timing recommendations?
- What counterfactual or matched-period methods are used to isolate effects from weather, construction and seasonal variation?
- Why prefer engineer-reviewed recommendations to autonomous signal execution, and what are the integration/safety constraints?

Databricks

- What are the measured storage, streaming ingestion and geospatial join costs for high-rate GPS/detector telemetry?
- How are traffic-state features versioned and monitored; can analysts update model features without bypassing signal engineering approval?

Tesla / vehicle-side sensing

- How do camera-based signal recognition and direct infrastructure-to-vehicle signal messages complement one another?
- How are recurring perception errors at particular intersections detected, triaged and audited?

13. Findings, unanswered questions and next research needs

Supported synthesis: Traffic signal coordination has multiple mature and experimental approaches; finite queue storage and sensing uncertainty are central engineering issues; AI benefits must be evaluated against strong baselines; operational safety requires a layer independent of policy learning.

Unresolved empirical questions: Which controller wins under the same four-intersection demand; how large is performance degradation under biased sampling; how frequently do controllers create spillback; whether observed differences are robust to held-out scenarios; and whether any benefit warrants additional deployment cost.

Next evidence needs: primary Jeju/Busan evaluation reports, detailed Boston evaluation protocol, field turning counts and travel times, sensor cost and reliability estimates, and operational constraints for any intended real intersection.

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